Create a Figure containing four subplots using add_subplot(). Plot (y=x), (y=x^2), (y=x^3), and (y=sqrt(x)).

```
import numpy as np
import matplotlib.pyplot as plt

# Create a Figure
fig = plt.figure()

# Create four subplots
ax1 = fig.add_subplot(2, 2, 1)
ax2 = fig.add_subplot(2, 2, 2)
ax3 = fig.add_subplot(2, 2, 3)
ax4 = fig.add_subplot(2, 2, 4)

# Generate x values
x = np.linspace(0, 5, 100)

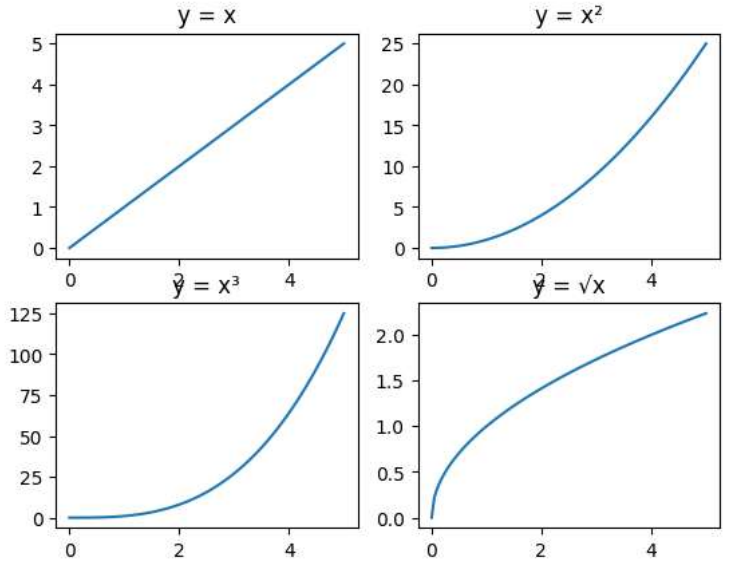
# Plot 1
ax1.plot(x, x)
ax1.set_title("y = x")

# Plot 2
ax2.plot(x, x**2)
ax2.set_title("y = x²")

# Plot 3
ax3.plot(x, x**3)
ax3.set_title("y = x³")

# Plot 4
ax4.plot(x, np.sqrt(x))
ax4.set_title("y = √x")

# Display
#plt.tight_layout()
plt.show()
```



usage of tight_layout()

```
import numpy as np
import matplotlib.pyplot as plt

# Create a Figure
fig = plt.figure()

# Create four subplots
ax1 = fig.add_subplot(2, 2, 1)
ax2 = fig.add_subplot(2, 2, 2)
ax3 = fig.add_subplot(2, 2, 3)
ax4 = fig.add_subplot(2, 2, 4)
```

```
# Generate x values
x = np.linspace(0, 5, 100)

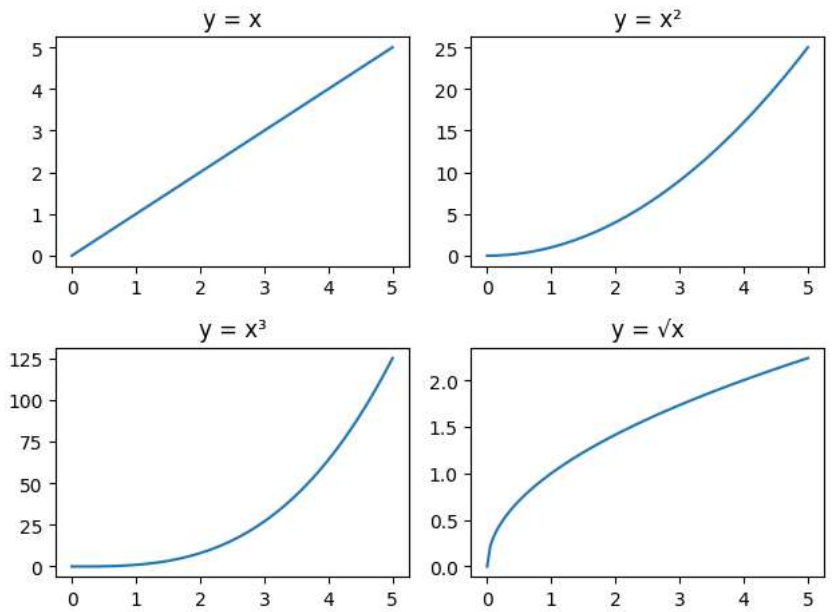
# Plot 1
ax1.plot(x, x)
ax1.set_title("y = x")

# Plot 2
ax2.plot(x, x**2)
ax2.set_title("y = x²")

# Plot 3
ax3.plot(x, x**3)
ax3.set_title("y = x³")

# Plot 4
ax4.plot(x, np.sqrt(x))
ax4.set_title("y = √x")

# Display
plt.tight_layout()
plt.show()
```



Create and position an Axes manually using add_axes(). Plot y=sin(x)

```
import numpy as np
import matplotlib.pyplot as plt

# Create Figure
fig = plt.figure()

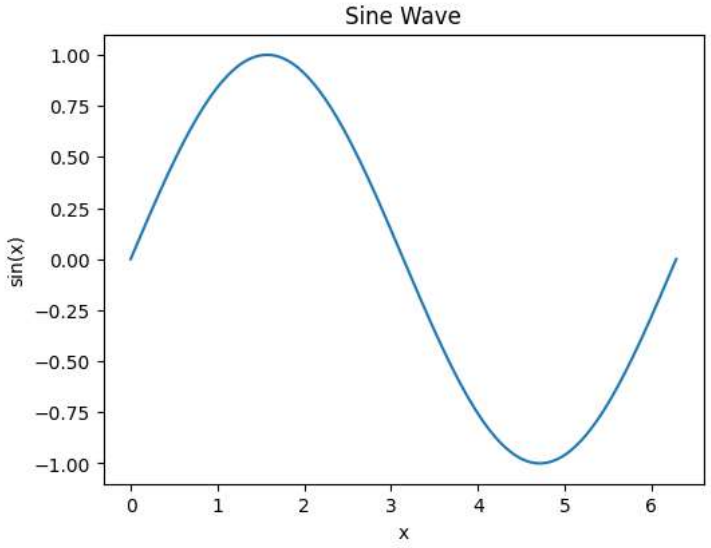
# Create Axes manually
ax = fig.add_axes([0.15, 0.15, 0.7, 0.7])

# Generate data
x = np.linspace(0, 2*np.pi, 200)
y = np.sin(x)

# Plot
ax.plot(x, y)

# Labels
ax.set_xlabel("x")
ax.set_ylabel("sin(x)")
ax.set_title("Sine Wave")

# Display
plt.show()
```



Create four plots using subplots() and explain how this differs from using add_subplot().

```
import numpy as np
import matplotlib.pyplot as plt

# Create 2 × 2 subplots
fig, ax = plt.subplots(2, 2)

x = np.linspace(0, 5, 100)

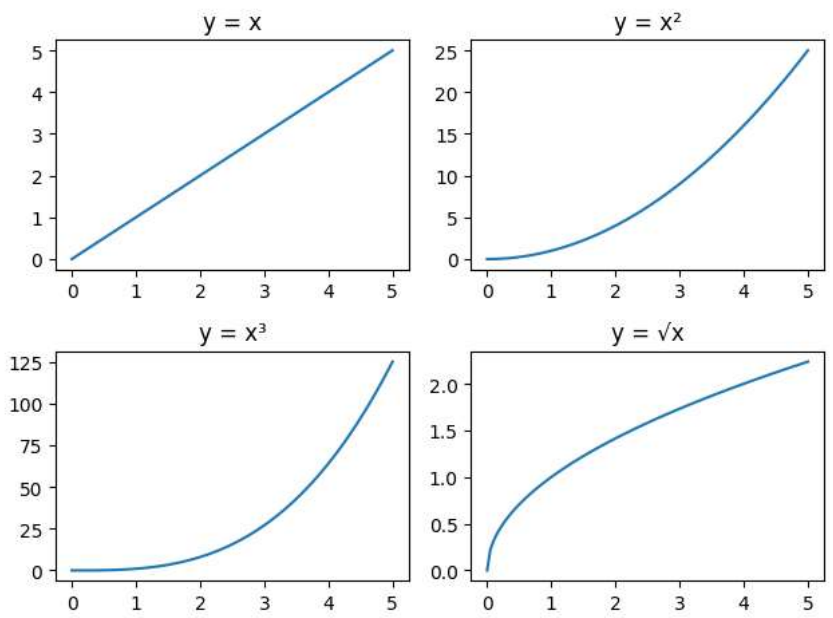
# Plot 1
ax[0, 0].plot(x, x)
ax[0, 0].set_title("y = x")

# Plot 2
ax[0, 1].plot(x, x**2)
ax[0, 1].set_title("y = x²")

# Plot 3
ax[1, 0].plot(x, x**3)
ax[1, 0].set_title("y = x³")

# Plot 4
ax[1, 1].plot(x, np.sqrt(x))
ax[1, 1].set_title("y = √x")

plt.tight_layout()
plt.show()
```



Listing 4.13 from Chapter 4, Section 4.1.7, “Subplots” demonstrates how four mathematical functions can be displayed in a 2 × 2 arrangement using plt.subplots(2,2). The book explains that the ax object behaves like a 2 × 2 NumPy array, so indexing determines the position of each subplot.

```
#13_subplot_functions.py
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 10, 100)
y1 = x
y2 = 5 - x
y3 = x**2
y4 = 1/(0.2*x + 1)

fig, ax = plt.subplots(2, 2)

# 1st row, 1st column
ax[0,0].set(ylabel='y', title='Linear function')
ax[0,0].plot(x, y1, 'b', lw=2)
ax[0,0].grid(True)

# 1st row, 2nd column
ax[0,1].set(title='negative slope')
ax[0,1].plot(x, y2, 'r', lw=2)
ax[0,1].grid(True)

# 2nd row, 1st column
ax[1,0].set(xlabel='x', ylabel='y', title='Parabola')
ax[1,0].plot(x, y3, 'g', lw=2)
ax[1,0].grid(True)

# 2nd row, 2nd column
ax[1,1].set(xlabel='x', title='Hyperbola')
ax[1,1].plot(x, y4, 'k', lw=2)
ax[1,1].grid(True)

fig.tight_layout()
plt.show()
```

